

8 Puzzle Problem using Iterative Deepening Search (IDS)

```
goal_state = (1, 2, 3,
              4, 5, 6,
              7, 8, 0)
print("Enter the initial state:")
initial_state = tuple(map(int, input().split()))
def display(state):
    for i in range(0, 9, 3):
        print(state[i], state[i + 1], state[i + 2])
    print()
def generate_moves(state):
    moves = []
    blank = state.index(0)
    row = blank // 3
    col = blank % 3
    # Move Up
    if row > 0:
        new_state = list(state)
        new_state[blank], new_state[blank - 3] = new_state[blank - 3], new_state[blank]
        moves.append(tuple(new_state))
    # Move Down
    if row < 2:
        new_state = list(state)
        new_state[blank], new_state[blank + 3] = new_state[blank + 3], new_state[blank]
        moves.append(tuple(new_state))
    # Move Left
    if col > 0:
        new_state = list(state)
        new_state[blank], new_state[blank - 1] = new_state[blank - 1], new_state[blank]
        moves.append(tuple(new_state))
    # Move Right
    if col < 2:
        new_state = list(state)
        new_state[blank], new_state[blank + 1] = new_state[blank + 1], new_state[blank]
        moves.append(tuple(new_state))
    return moves
```

```

def depth_limited_search(state, depth, path, states):
    states.append(state)
    if state == goal_state:
        return True
    if depth == 0:
        return False
    for next_state in generate_moves(state):
        if next_state not in path:
            path.add(next_state)
            if depth_limited_search(next_state, depth - 1, path, states):
                return True
            path.remove(next_state)
    return False

def ids():
    depth = 0
    while True:
        print("Searching at depth:", depth)
        print("States explored at this depth:")
        path = {initial_state}
        states = []
        if depth_limited_search(initial_state, depth, path, states):
            for i, state in enumerate(states):
                print("\nState", i + 1)
                display(state)
            print("Goal State Reached!")
            return
        for i, state in enumerate(states):
            print("\nState", i + 1)
            display(state)
        depth = depth + 1
    print("\nInitial State:")
    display(initial_state)
    print("Goal State:")
    display(goal_state)
    print("Applying Iterative Deepening Search...\n")
    ids()

```

```
===== RESTART: C:/Users/jainam/Desktop/AI LAB/IDS.py =====  
Enter the initial state:  
1 2 3 4 5 6 0 7 8  
  
Initial State:  
1 2 3  
4 5 6  
0 7 8  
  
Goal State:  
1 2 3  
4 5 6  
7 8 0  
  
Applying Iterative Deepening Search...  
  
Searching at depth: 0  
States explored at this depth:  
  
State 1  
1 2 3  
4 5 6  
0 7 8  
  
Searching at depth: 1  
States explored at this depth:  
  
State 1  
1 2 3  
4 5 6  
0 7 8  
  
State 2  
1 2 3  
0 5 6  
4 7 8  
  
State 3  
1 2 3
```

State 3

1 2 3
4 5 6
7 0 8

Searching at depth: 2

States explored at this depth:

State 1

1 2 3
4 5 6
0 7 8

State 2

1 2 3
0 5 6
4 7 8

State 3

0 2 3
1 5 6
4 7 8

State 4

1 2 3
5 0 6
4 7 8

State 5

1 2 3
4 5 6
7 0 8

State 6

1 2 3

7 0 8

State 6

1 2 3
4 0 6
7 5 8

State 7

1 2 3
4 5 6
7 8 0

Goal State Reached!